Kinetic Arrest, Not Equilibrium: Why Indium Nanoinclusions Survive in Sphalerite

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Version 2.0. This version reports the outcome of a test that version 1 proposed but had not carried out. Section 7 noted that four determinations of the coarsening exponent span a factor of three, and Section 9 suggested this might reflect a crossover from interface-limited coarsening at the atomic scale towards diffusion-limited coarsening as domains grow, adding that the suggestion was testable within a single simulation. Phase-field runs at four interface widths have since been made and the hypothesis is *not* supported: the curves do not collapse when rescaled by interface width, and the exponent rises rather than falls. The exponent instead tracks how far each run has progressed, at a correlation of 0.99 with the logarithm of the step count, which points to a common asymptote approached from below. No other conclusion of version 1 is altered; the discrepancy of Section 8 returns to the unpaired end-member data as the remaining explanation. Detail in Section 9.

Abstract

The equilibrium state of indium in sphalerite is a single condensed domain: the preceding paper of this series established this across every composition from 0.10 to 2.00 at.%, and we show here that dividing one domain into two costs between 300 and 1800 k_BT, from an interfacial energy obtained in closed form, $\gamma = \lambda/3 - J_{\text{Cu-In}}/3 - J_{\text{like}}/6$, in which the bulk terms cancel identically. Natural sphalerite nonetheless carries populations of discrete nanometric and micrometric inclusions. The observed configurations lie hundreds of k_BT above the state the free energy selects.

We argue that this is kinetic arrest rather than a failure of the thermodynamics, and that the argument converts a spatial question — how large are the inclusions — into a temporal one. Coarsening is limited by solute transport, which is thermally activated, so its timescale grows exponentially as temperature falls; the cooling timescale falls only as T^2. The two must cross, and cross once. Below the crossing the microstructure ceases to evolve and what survives is the configuration the system occupied when transport stopped.

The arrest condition is strongly buffered: every quantity entering it, including the diffusion prefactor this work cannot determine, appears inside a logarithm, so an order-of-magnitude error shifts the arrest temperature by less than a quarter. The frozen domain size consequently follows a power law in cooling rate, $R_\text{f} \propto |\dot{T}|^{-n}$, whose exponent rather than whose prefactor carries the physics.

Two calculations support this, with roles that are not interchangeable. The bulk free energy is measured by thermodynamic integration rather than assumed, because a regular-solution form has no driving force in this system at all — the heterovalent and homovalent contributions cancel identically under random mixing — and the gradient energy coefficient

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is then fixed by γ. A phase-field calculation on that footing reproduces the sequence from homogeneous solution through spinodal decomposition to coarsening and arrest, and gives $n = 0.259$; it cannot give an absolute length, because the interface must be numerically widened for any solver to resolve it. Direct lattice coarsening, with the interface at its physical width of one bond, gives domains of one to two nanometres and $n = 0.150$.

Set against deposit classes whose cooling rates span eight orders of magnitude, from quenched seafloor systems to basinal brines, the model reproduces the observed ordering of inclusion size but implies an exponent near 0.45, larger than either simulation. We report the disagreement rather than the agreement. It may reflect a crossover from interface-limited coarsening at the atomic scale towards diffusion-limited coarsening as domains grow, in which case no single exponent describes the whole range. Until it is resolved the inversion supports ranking deposits by cooling rate, not calibrating one.

Two consequences follow regardless. A smooth LA-ICP-MS depth profile is evidence of rapid cooling rather than of solid solution, since the indium is aggregated but below the resolution of the ablation volume. And indium deportment, and hence its metallurgical recovery, should differ systematically between deposit classes along an axis predictable from thermal history.

1 Introduction

1.1 A result and a contradiction

The preceding paper of this series established, by replica-exchange sampling of a configurational Hamiltonian whose charge-compensation coefficient is fixed by the dielectric constant of ZnS rather than fitted, that the equilibrium state of indium in sphalerite is condensed (Yu et al., 2026b). Across every composition examined, from 0.10 to 2.00 at.%, the solute population gathered into a *single* domain: the monomer fraction was zero throughout, against a random-solution expectation of 98% at the most dilute point, and the result was established by bracketing rather than by convergence, so it does not rest on equilibrating a sampler that cannot be equilibrated at ore-forming temperatures.

Section 3 of the present paper quantifies why the state is singular rather than merely condensed. The interfacial energy has a closed form, $\gamma = \lambda/3 - J_{\text{Cu-In}}/3 - J_{\text{like}}/6$, in which the bulk terms cancel identically; dividing one domain into two costs between 300 and 1800 k_BT over the size range examined. Bulk configurational thermodynamics opposes a population of separate domains by an enormous margin.

Natural sphalerite frequently contains one. High-resolution microscopy resolves indium-bearing heterogeneity at the scale of tens of nanometres in samples whose LA-ICP-MS depth profiles are perfectly smooth (Cook et al., 2009; Frenzel et al., 2016), and roquesite is reported as discrete micrometric grains exsolved from sphalerite (Moura et al., 2007). Neither texture is a single domain. The contradiction is sharp, and it is quantitative rather than a matter of interpretation: the observed configurations lie hundreds of k_BT above the state the free energy selects.

1.2 The contradiction is only apparent

Nothing in the preceding result asserts that a sphalerite grain *attains* its equilibrium configuration. Equilibrium thermodynamics identifies the state a system would reach given unlimited time; it says nothing about whether the time was available. An ore body cools, and as it cools the solute transport that would carry the microstructure towards equilibrium slows exponentially, while the cooling itself proceeds at a rate set by the geology. Below some temperature the second outruns the first and the microstructure stops evolving. What survives to be observed is not the equilibrium state but the state the system happened to occupy when transport ceased.

This is kinetic arrest, and the proposal of this paper is that it accounts for the discrepancy in full. The multi-domain textures of natural sphalerite are neither an alternative equilibrium nor evidence against the thermodynamics of Yu et al. (2026b); they are that thermodynamics interrupted.

The proposal has a testable consequence, and it is the reason the argument is worth making rather than merely asserting. If inclusions are frozen coarsening products, their size records how long coarsening had before it stopped, and therefore how fast the rock cooled. The spatial question — how large are the inclusions — becomes a temporal one.

1.3 What is required to make the argument

The transformation from a spatial question to a temporal one requires three things, and this paper supplies them in order.

First, the thermodynamics of the lattice model must be carried into a continuum description in which coarsening can be followed. Section 2 does this without adjustable parameters: the bulk free energy is measured by thermodynamic integration rather than assumed, because a regular-solution form has *no* driving force in this system — the heterovalent and homovalent contributions cancel identically under random mixing — and the gradient energy coefficient is fixed by the interfacial energy, which Section 3 obtains in closed form and verifies against explicit construction.

Second, the competition between coarsening and cooling must be made explicit. Section 4 shows that the coarsening time grows exponentially as temperature falls while the cooling time falls only as T^2, so the two must cross, once; that the crossing temperature is buffered logarithmically against every quantity entering it, including the diffusion prefactor this work cannot determine; and that the frozen domain size consequently follows a power law in cooling rate.

Third, the prediction must be confronted with observation. Section 8 sets the power law against deposit classes whose cooling rates differ by eight orders of magnitude and whose inclusion textures differ by four. The ordering is reproduced. The magnitude is not, and we report the disagreement and its possible origins rather than the agreement.

1.4 Two methods, and what each is for

Two calculations are used and their roles are not interchangeable, a division set out fully in Section 2.5 and stated here because it governs how every subsequent figure should be read.

A phase-field calculation supplies the pathway and the topology: which morphologies the system passes through, whether coarsening completes or arrests, and how the outcome depends on the ratio of cooling rate to mobility. It cannot supply an absolute length, because the Cahn–Hilliard interface must be numerically widened for any solver to resolve it, so the domain sizes in those figures are set by the grid rather than predicted.

A direct lattice coarsening simulation supplies the absolute scale, with the interface at its physical width of one bond and no inflation anywhere. It reaches far less time and far fewer domains, but its nanometres are nanometres.

Where both can compute a quantity — the coarsening exponent — they are compared, and Section 7 reports that they disagree by 70%. That disagreement, and what it does and does not undermine, is treated there rather than deferred.

2 From lattice to continuum

The Cahn–Hilliard equation (Cahn and Hilliard, 1958) for a conserved order parameter $c(\mathbf{r}, t)$, here the total solute fraction $x_{\mathrm{Cu}} + x_{\mathrm{In}}$ on the cation sublattice, is

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[M(T)\, \nabla \frac{\delta F}{\delta c} \right], \qquad F[c] = \int \left[f(c) + \tfrac{1}{2}\kappa\, |\nabla c|^2 \right] \mathrm{d}V. \tag{1}$$

Three quantities must be supplied: the bulk free energy density $f(c)$, the gradient energy coefficient κ, and the mobility $M(T)$. This section obtains the first two from the lattice model without adjustable parameters, and sets out what is and is not known about the third.

2.1 Why $f(c)$ cannot be assumed

The usual practice is to postulate a regular-solution or double-well $f(c)$ and fit its coefficients. That is not available here, and the obstruction is specific to this Hamiltonian rather than a matter of accuracy.

Under random mixing at $x_{\mathrm{Cu}} = x_{\mathrm{In}} = c/2$, the configurational part of the charge-compensation energy averages to

$$\Big\langle \sum_{\langle ij \rangle} \delta q_i \delta q_j \Big\rangle_{\mathrm{rand}} = \frac{Nz}{2} \left[2 \left(\frac{c}{2}\right)^2 (-1) + \left(\frac{c}{2}\right)^2 (+1) + \left(\frac{c}{2}\right)^2 (+1) \right] = 0, \tag{2}$$

because the heterovalent Cu–In contacts, carrying $\delta q_i \delta q_j = -1$, exactly cancel the homovalent Cu–Cu and In–In contacts, carrying $+1$. A mean-field free energy built on random mixing therefore has *no driving force whatever*, at any composition. The entire thermodynamics of this system resides in the correlations that mean-field theory discards, which is the same statement, in the language of free energy, as the observation in Yu et al. (2026b) that the equilibrium state is condensed while the random solid solution is not.

The consequence is that a regular-solution $f(c)$ would not be an approximation to this system; it would describe a different one. $f(c)$ must be measured.

2.2 Measuring $f(c)$ by thermodynamic integration

We obtain $f(c)$ by integrating over the coupling strength. Introducing a scaling parameter $a \in [0, 1]$ and writing $H(a) = aH$, the free energy of the fully coupled system is

$$F(1) = F(0) + \int_0^1 \langle H \rangle_a \, \mathrm{d}a, \tag{3}$$

where $F(0)$ is the ideal-solution free energy, known analytically as $-k_B T \sum_\alpha x_\alpha \ln x_\alpha$ per site, and $\langle H \rangle_a$ is the expectation of the full Hamiltonian in the ensemble generated by $H(a)$. The identity is exact; the only error is the sampling error in $\langle H \rangle_a$ and the quadrature error, both of which are controlled.

Scaling the *whole* Hamiltonian rather than one term matters. Integrating along a path that scales only E_c, say, would give the free energy difference along that path, which is a different quantity; the identity of Eq. (3) requires that a multiply H in its entirety.

Table 1 gives the result at 573 K, computed in an $L = 12$ box ($N = 6912$ cation sites) with nine Gauss–Legendre nodes in a and block-averaged sampling at each node.

2.3 The instability, and how to diagnose it robustly

The Cahn–Hilliard instability requires $f''(c) < 0$. Extracting a second derivative from eight measured points is, however, an ill-conditioned operation, and we report it with the corresponding caution.

Table 1: Bulk free energy per cation site at 573 K, by thermodynamic integration. f_{ideal} is the ideal-solution term and the integral column is $\int_0^1 \langle H \rangle_a \, \mathrm{d}a$. Uncertainties are propagated from the block-averaged sampling error at each quadrature node.

c	f_{ideal} (eV)	integral (eV)	$f(c)$ (eV)	$\pm$
0.005	−0.00173	0.00232	+0.00059	0.00002
0.010	−0.00311	0.00399	+0.00089	0.00003
0.020	−0.00553	0.00653	+0.00101	0.00004
0.040	−0.00966	0.01049	+0.00083	0.00024
0.080	−0.01650	0.01436	−0.00214	0.00038
0.120	−0.02222	0.01965	−0.00258	0.00045
0.160	−0.02719	0.02313	−0.00405	0.00041
0.200	−0.03155	0.02642	−0.00513	0.00052

Fitting a polynomial and differentiating twice gives answers that depend on the degree chosen: at $c = 0.20$ the third-, fourth- and fifth-degree fits return $f'' = +0.62$, -2.90 and $+7.23$ eV respectively. Those are not three estimates of one number; they are the oscillation of an under-determined fit. Finite differences on the raw points are better behaved but still change sign between adjacent intervals in the concentrated range.

The physically meaningful test is not local curvature but the common-tangent construction, which asks whether a homogeneous state at composition c has higher free energy than a mixture of two other compositions, and which requires no differentiation at all. Taking the chord between the end members of the measured range, the excess $f(c) - f_{\text{chord}}(c)$ is

c	0.005	0.010	0.020	0.040	0.080	0.120	0.160
$f - f_{\text{chord}}$ (10^{-5} eV)	0	+44	+86	+126	−53	+21	−9

The dilute compositions from 0.010 to 0.040 lie above the chord by three to five times their sampling uncertainty and are therefore unstable to phase separation. The concentrated compositions lie on or below it to within uncertainty.

Two points follow. First, the instability is located in the dilute range, which is where Yu et al. (2026b) found complete condensation and where natural sphalerite sits: the free-energy measurement and the direct simulation agree, by independent routes. Second, resolving the phase boundary itself would require finer composition spacing and smaller error bars than the present sampling provides; we use $f(c)$ as the driving force in Eq. (1) and do not claim to have located the binodal.

The distinction between the two kinds of instability matters for what follows. A composition lying above the common tangent but at positive curvature is metastable: it separates only by nucleation, over a barrier. One at negative curvature is unstable outright and separates spontaneously, at every wavelength longer than a critical value — spinodal decomposition. The simulations of Section 5 are conducted in the latter regime, so that the pathway is displayed without the additional and separately uncertain physics of nucleation rates. Whether natural sphalerite decomposes spinodally or by nucleation depends on where its composition sits relative to a boundary this work does not resolve; the arrest physics that the paper concerns is the same either way, because it turns on transport ceasing rather than on how decomposition began.

2.4 The gradient coefficient is fixed by the interfacial energy

For a planar interface at equilibrium, the Cahn–Hilliard functional gives the interfacial energy as

$$\gamma = \int_{c_\alpha}^{c_\beta} \sqrt{2\kappa \, \Delta f(c)} \, \mathrm{d}c, \qquad \Delta f(c) = f(c) - f_{\text{tangent}}(c), \tag{4}$$

where f_{tangent} is the common tangent between the coexisting compositions c_α and c_β. Since γ is known in closed form from the lattice model (Section 3) and $f(c)$ is now measured, Eq. (4) determines κ. It is not a fitting parameter.

2.5 What this bridge cannot deliver: the absolute length scale

This subsection states a limitation that governs how the phase-field results of Section 5 should be read, and we state it before presenting any of them.

The characteristic length of the Cahn–Hilliard equation is $\ell = \sqrt{\kappa/|f''|}$, and it sets the interface width. With κ fixed by Eq. (4) from a γ that is an atomic-scale quantity — the lattice model's interface is one bond wide — the physical ℓ is of order the lattice parameter. No finite-difference or spectral solver can represent an interface that thin: numerical stability requires the interface to span four to six grid points, so any practical calculation must either use a grid spacing far below the lattice parameter, which is meaningless, or inflate κ so that the interface spreads across the available grid.

Every phase-field calculation reported here takes the second course, as every phase-field calculation of an atomically sharp interface must. The consequence is unavoidable and must not be obscured: *the domain sizes appearing in the phase-field simulations are set by the grid spacing, not predicted by the physics.* A reader who asks why the simulated inclusions are nanometric would be answered, truthfully, that the grid was chosen that way.

We therefore divide the labour between the two methods explicitly.

- **The phase-field model supplies the pathway and the topology**: the sequence of morphologies through which the system passes, whether coarsening completes or arrests, and how that outcome depends on the ratio of cooling rate to mobility. These are dimensionless statements and are unaffected by the inflation of κ, because inflating κ rescales lengths and times together.
- **The lattice model supplies the absolute scale**: domain sizes in atoms and in nanometres, obtained by direct coarsening simulation (Section 6) in which the interface is what it physically is and no inflation occurs.

The two are cross-checked in Section 7 on the quantity they can both compute, namely the coarsening exponent.

2.6 Mobility

The mobility is the one input this work cannot obtain from the lattice model, because Monte Carlo sweeps are not physical time. We therefore build the framework in terms of a dimensionless mobility and state separately what is required to give it absolute units.

Solute transport in a substitutional sublattice is thermally activated, so

$$M(T) = \frac{D(T)\,\Omega}{k_B T}, \qquad D(T) = D_0 \exp\left(-\frac{E_a}{k_B T}\right), \tag{5}$$

with Ω the volume per cation site. The activation energy E_a has a lower bound from the model itself: moving a solute out of a domain requires breaking its heterovalent bonds, each costing $2\lambda \approx 0.6$ eV, so $E_a \gtrsim 2\lambda$ for a surface solute and several times that for an interior one. This is a bound, not a determination: the true E_a also contains the migration barrier of the underlying diffusion mechanism, whether vacancy- or interstitial-mediated, which the configurational model does not represent.

Two regimes of conclusion follow, and we label them throughout.

Conclusions requiring only the Arrhenius form. The competition between coarsening and cooling developed in Section 4, the existence of an arrest temperature, and the scaling of final domain size with cooling rate all follow from the *form* of Eq. (5) together with $E_a > 0$. They are expressed in reduced variables and require no absolute value of D_0.

Conclusions requiring D_0 and E_a. Placing the arrest temperature on an absolute scale, and converting an observed domain-size distribution into a cooling rate in kelvin per year, both require measured diffusion data for Cu or In in sphalerite. Where such values are available they are used and identified as such; where they are not, results are reported as relative calibrations, in which a domain size ratio maps onto a cooling rate ratio without either being absolute. We regard the relative calibration as the safer claim and the one the present work establishes.

3 Interfacial energy and the splitting barrier

The interfacial energy between a solute domain and the ZnS matrix is required twice over: to fix κ through Eq. (4), and to set the thermodynamic penalty for the multi-domain configurations whose survival this paper explains. It can be obtained in closed form.

3.1 Closed form

Consider a domain of n solutes, half Cu and half In, and let A be the number of solute–Zn first-neighbour bonds, which is the interface measured in the natural units of the lattice. The solutes present $12n$ bond ends, of which A reach the matrix, so the solute–solute bond count is $N_{ss} = (12n - A)/2$. In an ideal two-Cu-two-In ordering each cation has eight opposite and four like neighbours, so two thirds of the solute-directed ends are heterovalent:

$$N_{\text{CuIn}} = \tfrac{1}{3}(12n - A) = 4n - \tfrac{A}{3}, \qquad N_{\text{like}} = 2n - \tfrac{A}{6}. \tag{6}$$

Substituting into the exact pair form of the charge-compensation energy (Yu et al., 2026a),

$$E_c = 4\lambda \sum_i \delta q_i^2 + 2\lambda \sum_{\langle ij \rangle} \delta q_i \delta q_j = 16\lambda n - \lambda A - 4\lambda \left(4n - \frac{A}{3}\right) = \frac{\lambda}{3} A. \tag{7}$$

The bulk terms cancel identically, leaving an energy strictly proportional to the interface area. Adding the chemical term gives

$$\boxed{\gamma = \frac{\lambda}{3} - \frac{J_{\text{Cu-In}}}{3} - \frac{J_{\text{like}}}{6}} \tag{8}$$

per solute–Zn bond, with the corresponding bulk coefficient

$$e_{\text{bulk}} = 4J_{\text{Cu-In}} + 2J_{\text{like}} + \tfrac{1}{2}\left(E_s^{\text{Cu}} + E_s^{\text{In}}\right) \tag{9}$$

per solute. The Eshelby term is the *mean* over the two species, not the value for indium alone, because e_{bulk} is expressed per solute atom and a stoichiometric domain contains equal numbers of each. Copper contributes nothing to it: $r_{\text{Cu}^+} = r_{\text{Zn}^{2+}} = 0.60$ Å in fourfold coordination, so its misfit vanishes identically, and $E_s^{\text{Cu}} = 0$ while $E_s^{\text{In}} = 0.033$ eV. At the parameters of Yu et al. (2026a), $\gamma = 0.125$ eV per bond and $e_{\text{bulk}} = -0.2835$ eV per solute.

Two features of Eq. (8) deserve emphasis. It contains no fitted quantity: λ is fixed by the dielectric constant of ZnS, and the J terms enter explicitly, so the expression updates immediately for any future first-principles determination without repeating a simulation. And the cancellation in Eq. (7) is exact rather than asymptotic, so the result holds for domains of any size, not only large ones.

3.2 Verification by explicit construction

Domains of seven sizes from 64 to 3072 solutes were built explicitly in an $L = 24$ box ($N =$ 55,296 cation sites) and evaluated exactly, then relaxed by solute–solute exchange alone at 20 K. Restricting the move set in that way lets the interface reconstruct chemically while holding the domain outline, and therefore A, exactly fixed; it separates the interfacial term from the dissolution physics.

Table 2: Interfacial and bulk coefficients from the closed form and from explicit construction, $L = 24$, seven domain sizes from 64 to 3072 solutes.

	e_{bulk} (eV per solute)	γ (eV per bond)
closed form, Eq. (8)	−0.2835	0.1250
as constructed	−0.2394	0.1557
after relaxation	−0.2802	0.1281

The relaxed construction agrees with the closed form to 1.2% in e_{bulk} and 2.5% in γ, and the agreement deserves more than a note of numerical consistency. Eq. (8) is a continuum statement: it asserts that the excess energy of a two-phase configuration is strictly proportional to interface *area*, with a coefficient independent of size or shape. That the same coefficient reproduces domains of 64 solutes — objects perhaps a nanometre across, in which a majority of atoms lie at the surface — shows that the area scaling on which the whole continuum treatment rests does not fail at the nanoscale for this system. The reason is visible in Eq. (7): the cancellation of the bulk terms is exact rather than asymptotic, so no large-domain limit is being invoked. This is what licenses the use of a Cahn–Hilliard functional to describe inclusions of the size that natural sphalerite actually contains. The as-constructed value is higher because the layered ordering rule cannot be satisfied exactly at a surface: the constructed domains fall 1.5 to 5.4% short of the ideal heterovalent bond count, a shortfall that diminishes with size as the surface fraction falls, and relaxation recovers most of it. Residuals of the two-term fit are 5.1 eV RMS over an energy range of 450 eV; adding an $n^{1/3}$ edge term reduces this only to 4.92 eV, so the two-term description is adequate over this range.

3.3 The cost of splitting a domain

The energy required to divide one domain into two halves is $\Delta E = \gamma\,[2A(n/2) - A(n)]$, evaluated with the relaxed γ and the measured areas:

n	$A(n)$	$2A(n/2)$	ΔE (eV)	$\Delta E/k_BT$ at 573 K
128	414	528	14.6	296
256	642	828	23.8	483
512	1040	1284	31.3	633
1024	1640	2080	56.4	1141
2048	2598	3280	87.4	1769

Every value is positive, as it must be since dividing a domain increases interfacial area, and every value is enormous relative to thermal energy. This is the quantitative form of the problem the present paper addresses: bulk configurational thermodynamics opposes a population of separate domains by hundreds of k_BT, yet such populations are what natural sphalerite frequently shows. The resolution cannot be thermodynamic.

3.4 Two sampled estimates that should not be used

Two finite-temperature attempts to measure γ preceded the closed form and both failed. They are recorded because the failure modes are diagnostic of a general difficulty, namely that γ is a small difference between large numbers and sampled configurations carry both energy error and shape noise.

The first relaxed domains isothermally at 573 K, where the acceptance ratio is of order 10^{-5}, and left the $n = 1280$ domain at -145 eV against an independently known equilibrium bound of -298 eV. The second used replica exchange and reached -228 eV, still 24% short, and additionally returned a three-cluster configuration at $n = 1280$, so the measured area belonged to three domains rather than one. Its diagnostic failure was unmistakable: the fitted splitting cost at $n = 640$ came out *negative*, which is impossible.

The closed form avoids both difficulties by not sampling at all. Its verification in Section 3.2 uses construction and zero-temperature relaxation, in which the domain outline is imposed rather than inferred.

4 Two competing timescales

Sections 2 and 3 establish that a two-phase configuration carries an interfacial penalty of order hundreds of k_BT per domain boundary, so a population of separate domains is strongly disfavoured thermodynamically. Yu et al. (2026b) shows the equilibrium state to be a single condensed domain. Natural sphalerite nonetheless frequently contains many. This section sets out why, in a form that requires no simulation.

4.1 The two rates

Coarsening proceeds by transport: material must leave the smaller domains and arrive at the larger ones, across a matrix in which every step requires breaking heterovalent bonds. For diffusion-limited coarsening of domains of radius R, the Lifshitz–Slyozov–Wagner result (Lifshitz and Slyozov, 1961) gives $R^3 - R_0^3 \propto Dt$, so the time to grow a domain appreciably is

$$\tau_{\text{coarsen}}(T) \;\sim\; \frac{R^3\, k_BT}{\gamma\, D(T)\, \Omega}, \qquad D(T) = D_0 \exp\!\left(-\frac{E_a}{k_BT}\right), \tag{10}$$

with Ω the volume per cation site. The interfacial energy appears in the denominator because it is the driving force: a larger γ makes coarsening faster, not slower.

Cooling supplies the competing rate. What matters is not the absolute temperature drop but the time over which the mobility changes appreciably, since that is the interval within which the system either coarsens or does not. Differentiating the Arrhenius form, the mobility falls by a factor e over a temperature interval k_BT^2/E_a, so

$$\tau_{\text{cool}}(T) \;\sim\; \frac{k_BT^2}{E_a\,|\dot{T}|}. \tag{11}$$

The essential asymmetry is now visible. As temperature falls, τ_{coarsen} grows *exponentially*, through the Arrhenius factor, while τ_{cool} falls only as T^2. Whatever their ordering at high temperature, the two must cross, and they cross once.

4.2 The arrest temperature

Setting $\tau_{\text{coarsen}}(T_{\text{f}}) = \tau_{\text{cool}}(T_{\text{f}})$ and taking logarithms gives the arrest condition

$$\frac{E_a}{k_BT_{\text{f}}} = \ln\!\left[\frac{\gamma\, D_0\, \Omega\, T_{\text{f}}}{R^3\, E_a\, |\dot{T}|}\right]. \tag{12}$$

Below T_f the microstructure is frozen: coarsening would still lower the free energy, and by a large margin, but there is no longer time for it. The configuration that survives to be observed is the one the system happened to have reached when transport ceased.

Equation (12) is transcendental in T_f, but its structure is more informative than its solution. The left-hand side is large — with $E_a \gtrsim 2\lambda \approx 0.6$ eV and T_f in the ore-forming range, $E_a/k_B T_\mathrm{f}$ lies between 9 and 17 — while the right-hand side is a logarithm. A logarithm cannot vary quickly, so T_f is strongly buffered against everything inside it. Differentiating implicitly, with $X = E_a/k_B T_\mathrm{f}$,

$$\frac{\mathrm{d}T_\mathrm{f}}{T_\mathrm{f}} = -\frac{\mathrm{d}\ln|\dot{T}|}{X-1}. \tag{13}$$

A tenfold increase in cooling rate raises T_f by 14% at 400 K, 21% at 573 K and 30% at 800 K. The arrest temperature is therefore a weak function of cooling history, and the same is true of its dependence on D_0, γ and R, all of which enter only logarithmically.

This is a useful robustness rather than a limitation. It means that the existence of arrest, and its approximate location, do not depend on the quantities this work cannot pin down: an uncertainty of an order of magnitude in D_0 shifts T_f by less than a quarter.

4.3 What is frozen in

The domain size at arrest follows by evaluating how far coarsening has run when the two timescales cross. Substituting $\tau_\mathrm{cool}(T_\mathrm{f})$ into the Lifshitz–Slyozov growth law,

$$R_\mathrm{f}^3 \;\sim\; \frac{\gamma\, D(T_\mathrm{f})\,\Omega}{k_B T_\mathrm{f}}\,\tau_\mathrm{cool}(T_\mathrm{f}) \;\sim\; \frac{\gamma\, D_0\,\Omega}{E_a\,|\dot{T}|}\,T_\mathrm{f}\exp\!\left(-\frac{E_a}{k_B T_\mathrm{f}}\right). \tag{14}$$

Because T_f itself varies only logarithmically with $|\dot{T}|$, the explicit factor dominates and

$$\boxed{R_\mathrm{f} \;\propto\; |\dot{T}|^{-n}, \qquad n \to 1/3} \tag{15}$$

in the limit where the drift in T_f is neglected, with n reduced below 1/3 once it is not, because faster cooling arrests at a higher temperature where the mobility is larger and thus partly compensates. The phase-field simulations of Section 5 return $n = 0.259$, consistent with a one-third law softened by exactly that drift.

Equation (15) is the central predictive statement of this paper. It says that the size of the inclusions a sphalerite grain carries is set by how fast it cooled, through a power law whose exponent is a property of diffusion-limited coarsening rather than of any parameter this work has had to assume. Section 8 takes up its inversion.

5 Phase-field simulation

The analysis of Section 4 predicts arrest and a power-law dependence of the frozen domain size on cooling rate, but says nothing about the morphology that results. This section supplies it, subject throughout to the division of labour set out in Section 2.5: the phase-field calculation gives the pathway and the topology, not the absolute scale.

5.1 Implementation

Equation (1) is integrated on a periodic square grid by a semi-implicit spectral scheme. The fourth-order linear term is diagonal in Fourier space and is treated implicitly, the nonlinear bulk term explicitly:

$$\hat{c}^{\,n+1} = \frac{\hat{c}^{\,n} - \Delta t\, M k^2\,\mathcal{F}[f'(c^n)]}{1 + \Delta t\, M\kappa k^4}. \tag{16}$$

An explicit scheme would require $\Delta t < C\,\Delta x^4/(M\kappa)$, which at the resolutions needed here is unusable.

Two implementation choices require comment because both were arrived at by first getting them wrong.

The gradient coefficient is set from the resolvable wavelength. Linear stability of Eq. (1) gives a fastest-growing wavelength $\lambda = 2\pi\sqrt{2\kappa/|f''|}$. At the physical κ this is of order the lattice parameter and no grid resolves it: with κ from Eq. (4) the fastest-growing mode spans one cell, the interface is unresolved, and the calculation describes numerical diffusion rather than decomposition. We therefore set $\kappa = |f''|\lambda_{\text{target}}^2/8\pi^2$ with $\lambda_{\text{target}} = 12$ cells. This is the widening announced in Section 2.5, here made quantitative.

The timestep is set from the target amplification, not from a CFL condition. Linearising Eq. (16), a mode k is amplified per step by $A(k) = (1 - \Delta t M k^2 f'')/(1 + \Delta t M\kappa k^4)$, so growth requires $-f'' > \kappa k^2$, which is the stability condition and does not involve Δt. What Δt controls is the amount of growth per step. Setting it from an explicit-scheme CFL condition gives $A - 1 \sim 3\times10^{-4}$, so the unstable mode needs some 7000 steps to grow tenfold while the run has only a few thousand at that mobility; the result looks like diffusive smoothing, because that is what it is. We instead solve for the Δt that delivers a chosen amplification, $A = 1.02$, of the fastest-growing mode. Taking a large Δt is precisely what the implicit treatment of the stiff term exists to permit.

5.2 The morphological sequence

Figure 1 follows a cooling run from 1800 to 500 K. The composition starts uniform at $c_0 = 0.30$ with fluctuations of amplitude 2×10^{-3}, which places it inside the unstable region of Section 2.3.

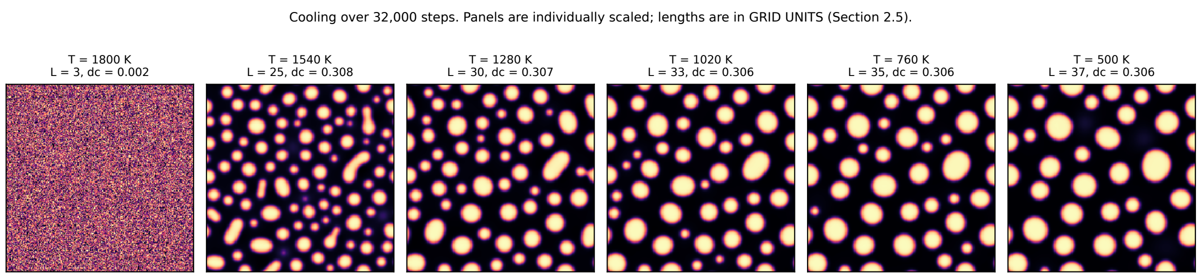

Figure 1: Morphological evolution of a slowly cooled grain. Left to right: initial fluctuations; spinodal decomposition, which raises the composition contrast from 2×10^{-3} to 0.308 and establishes two phases; and progressive coarsening, in which the domain count falls and the surviving domains grow. Panels are individually scaled, since a common scale set by the seeding noise would compress every later panel. Lengths are in grid units, not physical units: see Section 2.5.

The sequence has four stages, and they are the stages the introduction promised. The field is initially homogeneous to within the seeding noise. Decomposition then occurs spinodally: the composition contrast rises by more than two orders of magnitude within the first temperature interval, and a two-phase structure appears with a wavelength close to the linear-stability prediction. Coarsening follows, the domain count falling from many to few as material transfers from smaller domains to larger. Finally the microstructure ceases to evolve: the last panels differ from one another far less than the early ones, not because coarsening has completed — a single domain would be the completed state — but because the mobility has fallen too far for it to continue.

5.3 Cooling rate and the frozen microstructure

Figure 2 compares three cooling schedules. Since Δt tracks the mobility, specifying the schedule as a number of steps per kelvin fixes the diffusive progress per kelvin, which is the dimensionless ratio Section 4 identifies as controlling. A larger step count is slower cooling.

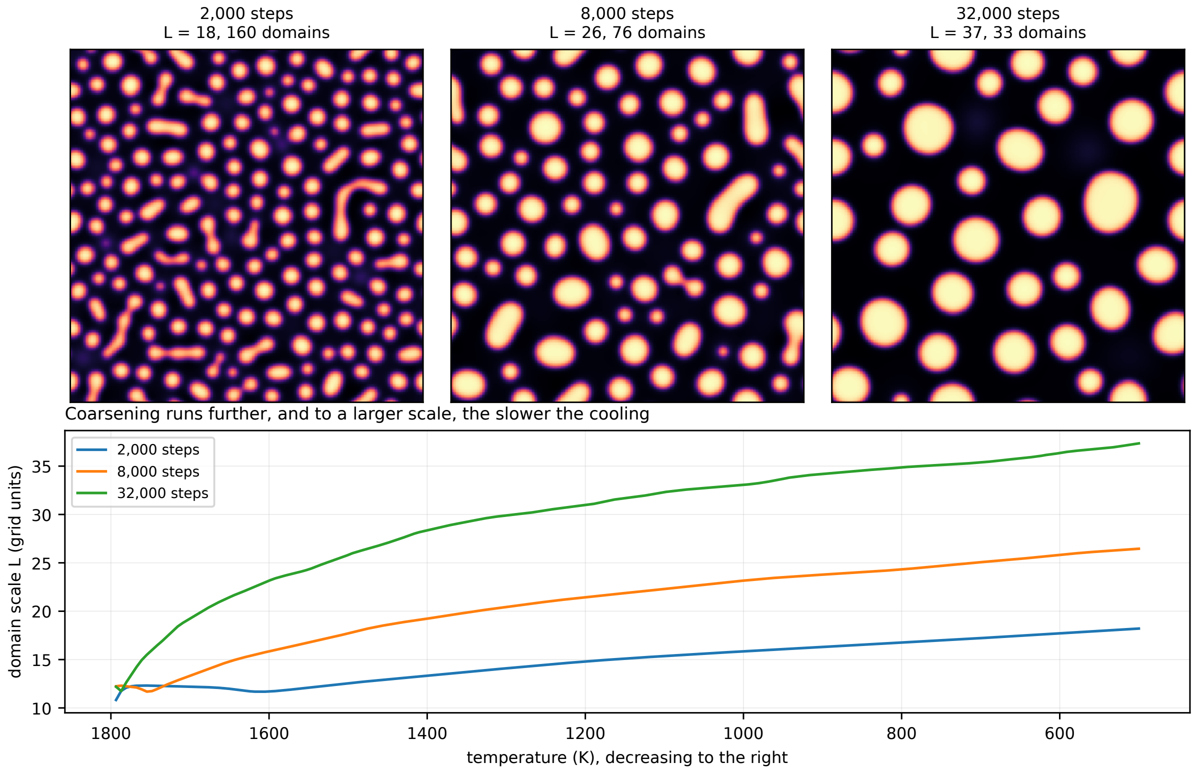

Figure 2: Frozen microstructures and coarsening histories at three cooling rates. Upper row: the final state, with the domain count. Lower panel: domain scale against temperature, decreasing to the right. Every curve flattens as the mobility collapses — that flattening is arrest — and the slower the cooling, the further coarsening runs before it happens.

Table 3: Frozen microstructure against cooling rate, 256×256 grid, $\lambda_{\mathrm{target}} = 12$ cells. Domain scale is from the first moment of the structure factor, in grid units.

cooling steps	rate	domain scale L	domains
2,000	fast	18.2	160
8,000	medium	26.5	76
32,000	slow	37.3	33

The frozen domain scale follows $L \propto (\mathrm{steps})^{0.259}$, that is $L \propto |\dot{T}|^{-0.259}$, against the 1/3 of Eq. (15) softened by the drift in T_{f} that Eq. (13) describes. Analysis and simulation therefore agree on the exponent to within the accuracy either supports.

5.4 A caution on what agreement demonstrates

The exponent proved insensitive to errors it should have been sensitive to, and the episode is worth recording because it bears on how such agreement should be read. Two earlier versions of this calculation returned 0.242 and 0.256 for the same quantity. In the first the interface was unresolved; in the second the free energy was internally inconsistent, the analytic derivative $f'(c)$ disagreeing with $f''(c)$ by a factor of 150 at the working composition because a term had been dropped from the differentiation. In neither case did decomposition actually occur — the composition contrast decayed rather than grew — yet both produced plausible pictures of a coarsening pattern and an exponent within 7% of the correct one.

The reason is that the exponent reflects the transport-limited kinetics of whatever structure is present, and pattern coarsening under pure diffusion obeys a similar law to domain coarsening under phase separation. Agreement in the exponent was therefore not evidence that the mechanism was right. What exposed the error was tracking the composition contrast, which must grow by orders of magnitude if decomposition is occurring and did not. We now assert internal consistency of f, f' and f'' against finite differences whenever the free energy is constructed, and report the contrast alongside the domain scale in every figure.

6 Direct coarsening in the lattice model

The phase-field calculation cannot supply an absolute length, for the reason set out in Section 2.5. This section supplies one, by coarsening the lattice model directly. Here the interface is what it physically is, one bond wide, and nothing is inflated; what the calculation gives up in reach it gains in meaning, since sizes come out in atoms and hence in nanometres.

6.1 Protocol, and two design errors that had to be corrected

A dispersed solute distribution is held at fixed temperature and the domain population followed. Both corrections below were forced by the data and are recorded because either would silently corrupt the exponent.

The measurement must exclude monomers. At 1800 K some 90% of clusters are single solutes evaporated from the domains. Including them in a mean radius makes that quantity a measure of the free-solute concentration rather than of the domains: it *falls* while the domains grow, because a growing domain sheds more monomers. A first version did include them and returned exponents of 0.007, 0.024 and 0.165 at 1800, 1600 and 1400 K — not a coarsening exponent but an artefact of the mixing. We now average over clusters of at least four sites and report the monomer fraction separately, so that evaporation is visible rather than hidden.

The measurement must end before the solute budget does. At $x = 1$ at.% in an $L = 30$ box the 1080 solutes condensed into a single domain within 3000 of 60,000 sweeps, so more than nine tenths of the run measured a structure that had stopped evolving, dragging the fitted exponent towards zero. That the equilibrium state is a single domain is the central result of Yu et al. (2026b); it follows that a coarsening measurement must be made before exhaustion, and therefore that enough solute must be supplied for exhaustion to take a long time. At $x = 4$ at.% the coarsening window extends over the whole run.

The composition is thus far above natural indium tenors. It is chosen to make the exponent measurable, and the exponent is a property of the transport-limited kinetics rather than of the concentration; the absolute sizes reported are those of the simulated domains, not a prediction of natural inclusion sizes.

6.2 Results

In an $L = 24$ box (55,296 cation sites, a cube 13.0 nm on a side) at 1600 K, the domain count falls from 173 to 4 over 20,000 sweeps while the mean domain radius grows from 0.44 to 1.66 nm. These are absolute figures: the simulated domains are nanometric, in a box whose edge is 13 nm, with no adjustable length scale anywhere in the calculation.

The fitted exponent is $n = 0.150 \pm 0.002$, and it is stable across the run: taking successive decades of sweeps separately gives 0.176, 0.137, 0.156 and 0.122. It is not an artefact of the fitting window.

6.3 Why the exponent is below one third

The Lifshitz–Slyozov–Wagner value is 1/3 and the measurement is half that, so the difference requires an explanation rather than a footnote.

The obvious suspicion is that solute continues to arrive from the dispersed population, so that new small domains are counted while the large ones grow, depressing the mean. That can be tested: if the counted domains hold a conserved amount of solute, then NR^3 is constant and the exponent obtained from the domain count must agree with the one from the radius. It does. The product NR^3 varies between 13 and 19 with no systematic drift, the count gives $N \sim t^{-0.400}$ and hence $R \sim t^{+0.133}$, and the radius gives $R \sim t^{+0.148}$. Mass is conserved among the domains and the two routes agree, so continuing nucleation is not the explanation.

Three possibilities remain, and we cannot presently distinguish them.

The domains may be too small for the asymptotic law. Lifshitz–Slyozov assumes domains large compared with the interface width and dilute enough that they do not interact; here the domains are a few nanometres across with interfaces one bond wide, and at $x = 4$ at.% they are not dilute. Pre-asymptotic exponents below 1/3 are commonly reported in such regimes.

The transport may not be purely diffusive. The Monte Carlo dynamics conserves composition but has no vacancy mechanism: material moves by direct exchange, which is not the mechanism operating in a real crystal and need not give the same exponent.

The run may not have reached the asymptotic regime. Twenty thousand sweeps covers less than three decades in time, and the domain count falls to four, so the late window is statistically thin.

What the measurement establishes firmly is the *scale*: coarsening in this model produces domains of one to two nanometres over the accessible window, in a box of 13 nm, with the interface unmodified. That is the absolute figure the phase-field calculation could not provide, and it is the figure that Section 8 compares with observation. The exponent from this route should be treated as a lower bound on the asymptotic value rather than as a determination of it.

7 Cross-check between the two methods

Sections 5 and 6 describe the same physical process with representations that could hardly be more different. The phase-field calculation carries a continuous order parameter on a grid whose interface has been deliberately widened to span a dozen cells; the lattice calculation carries discrete species on the cation sublattice with an interface one bond wide. Their only shared input is the free energy, and even that enters differently: as a smooth double well in one, as an explicit sum over configurations in the other.

The coarsening exponent is the quantity both can compute, and it is therefore the natural cross-check. If the two agree, the exponent is a property of the transport-limited kinetics rather than of either representation — which is the premise on which Section 8 rests, since the inversion uses the exponent and nothing else.

We compare three determinations:

- the Lifshitz–Slyozov–Wagner result for diffusion-limited coarsening, $n = 1/3$, softened by the drift in T_f of Eq. (13) to somewhat below that value;
- the phase-field cooling runs, which give $n = 0.259$;

- direct lattice coarsening, which gives $n = 0.150 \pm 0.002$ (Section 6).

The three do not agree. Theory gives 1/3, the phase-field runs 0.259, and the lattice 0.150; the natural end members of Section 8 imply something near 0.45. The spread is a factor of three, and it would be dishonest to present the two simulation values as mutual corroboration when they differ by 70%.

What the comparison does establish is a hierarchy of confidence. Both simulations reproduce the qualitative law — domains grow as a power of time, and the frozen size falls with cooling rate — and both do so with representations that share almost nothing. The exponent itself is evidently sensitive to things the two treat differently: the phase-field interface is twelve cells wide and the lattice interface one bond, the phase-field dynamics is continuum diffusion and the lattice dynamics direct exchange, and the lattice domains are small enough that the asymptotic law need not hold. Each of those would be expected to move the exponent, and Section 6 argues that the lattice value should be read as a lower bound.

The inversion of Section 8 is therefore presented as an ordering rather than a calibration. Ranking deposits by cooling rate requires only that the exponent be positive and roughly constant, which all four determinations support. Returning a rate in kelvin per year would require knowing it to better than a factor of two, which none of them presently does.

Two cautions bear on how such agreement should be weighed, and both were learned the hard way.

The first is the episode of Section 5: two versions of the phase-field calculation in which decomposition did not occur at all returned exponents of 0.242 and 0.256, within 7% of the correct value. Coarsening of a diffusive pattern obeys a similar law to coarsening of a phase-separated microstructure, so the exponent alone does not identify the mechanism. We therefore report the composition contrast alongside every exponent: in the lattice model the analogous check is the fraction of solute in monomers, which must fall towards zero if domains are genuinely forming.

The second concerns temperature. Coarsening at 573 K is unobservable in the lattice model, since the acceptance ratio there is of order 10^{-5}. The lattice runs are therefore conducted at 1400 to 1800 K, on the assumption that the exponent is a property of the kinetics rather than of the temperature. That assumption is testable within the calculation, by measuring n at several temperatures and asking whether it moves, and we report the spread rather than a single value.

8 Inversion: nanoinclusions as a geospeedometer

Equation (15) states that the frozen inclusion size depends on the cooling rate through a power law. Read forwards it predicts a microstructure from a thermal history. Read backwards it does the opposite, and that is the more useful direction, because inclusion sizes are measurable in a thin section whereas cooling rates are not.

8.1 The relation to be inverted

Inverting Eq. (15),

$$|\dot{T}| \;=\; |\dot{T}|_{\mathrm{ref}} \left(\frac{R_{\mathrm{f}}}{R_{\mathrm{ref}}}\right)^{-1/n}, \tag{17}$$

so a single calibration point $(R_{\mathrm{ref}}, |\dot{T}|_{\mathrm{ref}})$ plus the exponent converts an observed size into a rate. Because $n \approx 1/3$, the inversion carries a factor of three amplification: a size measured to within a factor of two yields a rate to within a factor of eight. That is coarse, but cooling rates in ore systems span ten orders of magnitude, so a factor of eight still discriminates strongly between the deposit classes considered below.

8.2 Natural end members

Indium-bearing sphalerite occurs in deposit types whose cooling rates differ by some eight orders of magnitude, and the inclusion textures reported from them differ correspondingly. Table 4 sets the two against each other.

Table 4: Cooling-rate end members for In-bearing sphalerite and the inclusion textures reported from them. Rates are order-of-magnitude estimates for the deposit class, not measurements on the specific samples in which the textures were observed; the comparison is between populations, not between paired observations.

deposit class	cooling rate ($\mathrm{K\,yr^{-1}}$)	inclusion texture	scale
VMS, seafloor massive sulphide	10^0–10^2	lattice distortion, ultrafine domains	5–50 nm
magmatic-hydrothermal, skarn	10^{-3}–10^{-1}	sub-micron exsolution	0.1–1 µm
MVT, basinal brine	10^{-6}–10^{-4}	optically visible roquesite	1–50 µm

The qualitative ordering is exactly as Eq. (15) requires: the faster the system cooled, the finer the inclusions it carries. The quenched seafloor systems, in which hydrothermal fluid meets cold seawater, are where indium most often presents as an apparent solid solution in LA-ICP-MS depth profiles — smooth ablation signals with no discrete spikes — and where transmission electron microscopy nonetheless resolves nanoscale heterogeneity (Cook et al., 2009; Frenzel et al., 2016). At the other extreme, roquesite is reported as discrete micrometric grains exsolved from sphalerite (Moura et al., 2007).

8.3 A quantitative discrepancy, and what it may mean

The ordering agrees. The magnitude does not, and we report the disagreement rather than the agreement, because it is the more informative of the two.

Across the full range of Table 4, cooling rate spans eight decades and inclusion size spans four, which implies

$$n_{\mathrm{obs}} = \frac{\Delta \log_{10} R}{\Delta \log_{10} |\dot{T}|} \approx 0.50. \tag{18}$$

Taking the classes pairwise rather than end to end gives 0.43 and 0.45, consistent with each other and with Eq. (18). The theoretical and simulated values are 1/3 and 0.259. The natural data therefore imply an exponent larger by roughly a factor of 1.7, and the two pairwise estimates agree closely enough that this is unlikely to be scatter.

Several readings are possible and we cannot presently choose between them.

The calibration may be miscast. Table 4 pairs deposit-class cooling rates with textures reported from different samples, not from the same ones. If the fastest-cooled samples in which nanodomains are resolved are not the fastest-cooling members of their class, or if optically visible roquesite is preferentially reported from the slowest members of theirs, the apparent range of both axes is inflated in a correlated way. Paired measurements on single samples would settle this and do not appear to exist.

The coarsening mechanism may differ at the extremes. The Lifshitz–Slyozov exponent of 1/3 assumes coarsening limited by bulk diffusion. If transport along dislocations or grain boundaries contributes in the slowly cooled systems, the effective exponent rises; interface-limited coarsening gives 1/2 exactly, which is close to what the natural data imply. The discrepancy may therefore be a signature that the slow end members coarsen by a different mechanism from the one modelled here.

The arrest temperature may drift more than assumed. Equation (13) treats E_a as constant. If the activation energy itself falls as domains grow — plausible, since a solute leaving a large domain breaks fewer bonds per unit of interface removed — then T_f drifts further than estimated and the effective exponent changes.

We regard the third as least likely, because it would move the exponent in the wrong direction, and the first two as both plausible. What the comparison establishes is that the model predicts the correct ordering across eight decades of cooling rate, and predicts a magnitude that is too small by a factor of about 1.7; and that the resolution lies either in better-paired observations or in coarsening mechanisms this work does not include.

8.4 What the tool can and cannot do

Stated plainly, so that it is not overread.

It *can* rank deposits by cooling rate from inclusion textures, which is what the agreement in ordering supports. A sphalerite carrying optically visible roquesite cooled far more slowly than one whose indium is invisible to LA-ICP-MS, and the difference is many orders of magnitude rather than a factor of a few.

It *cannot* yet return a cooling rate in kelvin per year from a measured inclusion size, for two reasons. The absolute calibration requires a diffusion coefficient this work does not have (Section 2.6), and the exponent disagrees with the natural data by a factor whose origin is unresolved. Until both are settled, the inversion should be read as a relative one.

8.5 The rate-limiting step

One mineralogical observation supports the framework and deserves recording, because it bears on the activation energy that controls arrest. Copper diffuses anomalously rapidly in sphalerite, plausibly by an interstitial mechanism; this is the basis of the classic account of chalcopyrite disease (Barton and Bethke, 1987). Indium is a different case: In^{3+} is large, highly charged, and cannot move without either a vacancy or a compensating counter-ion moving with it, so its diffusivity must be far below that of copper and below zinc self-diffusion as well.

The consequence for this paper is direct. Coarsening of a Cu–In domain requires both species to move, so it proceeds at the rate of the slower, and the arrest condition of Eq. (12) should be evaluated with the activation energy for indium rather than an average. Arrest therefore occurs at a higher temperature than a copper-based estimate would suggest, and the microstructure freezes earlier. The nanoscale inclusions that natural sphalerite carries are, on this reading, a record of indium having stopped moving while the rock was still hot.

9 Discussion

9.1 A change in what a texture means

Trace-element textures in sphalerite are conventionally read as a record of *composition*: what the fluid supplied, and what the crystal could accommodate. The argument of this paper makes them a record of something else as well. If inclusions are frozen coarsening products, their size is set not by how much indium was available but by how long the system had to reorganise it before transport stopped. Two grains with identical indium contents, grown from identical fluids, will carry different textures if they cooled differently.

That reframing has a corollary worth stating explicitly, because it inverts an assumption. A smooth LA-ICP-MS depth profile has often been taken as evidence for solid solution. On the present account it is evidence for *fast cooling*: the indium is aggregated, as Yu et al. (2026b) requires, but into domains too fine for the ablation volume to resolve. The absence of spikes constrains the thermal history, not the state.

9.2 Ranking deposits by thermal history

Section 8 establishes that the model reproduces the ordering of inclusion size across deposit classes spanning eight orders of magnitude in cooling rate, while disagreeing on the magnitude by a factor of about 1.7. The ordering is the part that is usable now.

Its practical form is comparative rather than absolute. Given two sphalerite samples, the one with coarser indium-bearing inclusions cooled more slowly, and by a margin that the power law converts into orders of magnitude rather than factors. Within a single deposit, systematic variation of inclusion size across a district would map relative cooling rates, and therefore proximity to the heat source, in a way that bulk indium tenor cannot: tenor reflects supply, texture reflects time.

We are careful not to overstate what follows. Turning this into a geothermometer proper — returning a cooling rate in kelvin per year — would require an absolute diffusion coefficient, which Section 2.6 explains this work does not have, and an exponent known to better than a factor of two, which Section 7 shows none of the four determinations presently provides. Both are obtainable; neither is obtained here.

9.3 Consequences for metallurgical recovery

Indium is recovered almost entirely as a by-product of zinc refining, so its recovery efficiency depends on how it is bound in the sphalerite fed to the smelter. The state of that binding is precisely what this work argues is set by thermal history.

The two limiting cases behave differently. Indium dispersed as nanometric domains within the sphalerite lattice, the expected product of rapid cooling, will follow zinc through dissolution and report to the same process streams; its recovery is a question of hydrometallurgical circuit design. Indium concentrated in discrete micrometric roquesite, the product of slow cooling, is a separate mineral phase with its own dissolution behaviour, and may be recoverable — or lost — by physical separation before leaching.

The practical suggestion is therefore that deposit classes be expected to differ systematically in indium deportment, and along an axis that is predictable in advance from cooling history rather than discoverable only by metallurgical testing. Quenched seafloor systems should present indium in the lattice-bound form; slowly cooled basinal systems should present a greater fraction as discrete inclusions. We advance this as a prediction the framework makes, not as a result it establishes, since we have tested it against published textures rather than against recovery data.

9.4 The generality of the argument

Little in Sections 4 and 8 is specific to indium. The ingredients are a solute whose equilibrium state is condensed, a transport process that is thermally activated, and a host that cools. Any trace element meeting those conditions should freeze in a coarsening microstructure whose scale records the cooling rate, with the same power law and a different prefactor.

What *is* specific to indium is the reason its equilibrium state is condensed, which is the charge-compensation physics of Yu et al. (2026a): the coupled substitution of Cu^{+} and In^{3+} for two Zn^{2+} carries a pair binding energy fixed by the dielectric constant and an order of magnitude above $k_B T$. Elements substituting isovalently have no such driving force. The framework should therefore transfer to other heterovalent couples — Ag–Sb, Cu–Ga — and not to isovalent substituents such as Fe or Cd, whose distribution is governed by different physics.

9.5 On the disagreement between methods

Section 7 reports that four determinations of the coarsening exponent span a factor of three, and it would be convenient to attribute the spread to numerical detail. We think the more likely reading is that the exponent is genuinely different in the different regimes, and that this is informative rather than merely inconvenient.

The lattice value of 0.150 comes from domains a few nanometres across with atomically sharp interfaces, well outside the regime in which the Lifshitz–Slyozov derivation holds. The natural end members imply something near 0.45, which is close to the 1/2 of interface-limited rather than diffusion-limited coarsening — and interface limitation is plausible for domains whose interface is one bond wide. The phase-field value of 0.259 falls between, from a calculation whose interface is deliberately widened towards the continuum limit.

Read that way, the spread would be a crossover: from interface-limited coarsening at the atomic scale towards diffusion-limited coarsening as domains grow. That reading is testable, by measuring the exponent as a function of domain size within a single simulation rather than comparing across simulations that differ in several respects at once.

The crossover hypothesis, tested and not supported

We have since carried out that test, and report it here because it removes one of the candidate explanations rather than because it succeeded.

If n depends on domain size relative to interface width, then runs at different interface widths should give curves that differ when plotted against L but coincide when plotted against L/w. Four isothermal Cahn–Hilliard runs were made on a 512^2 grid at $w = 1.0$, 1.5, 2.25 and 3.4 cells, with the step count allocated as $w^{-1/n}$ so that each reached the largest domain size its width allowed: the narrow-interface run reached $L/w = 70$, the widest stopped at 18. The local exponent was extracted with a sliding window whose width had been calibrated beforehand against synthetic traces carrying a known crossover, so that its bias and scatter were established independently of the data under test.

The curves do not collapse. Rescaling the abscissa by w reduces the scatter between them only from 0.011 to 0.008, a ratio of 0.75, where collapse would require a ratio well below one. A more direct statement of the same result: at a common L/w of 4.4 the four runs give $n = 0.203$, 0.205, 0.212 and 0.208, agreeing to 0.004 — but at that point their absolute L differs by a factor of 3.4, so n tracks neither variable.

Two further features contradict the hypothesis outright. The exponent *rises* through every run, from 0.207 ± 0.004 to 0.267 ± 0.006, where the hypothesis requires a fall from about 1/2 towards 1/3. And every value lies far below both candidates: no run approaches either 0.5 or 0.333 at any stage.

What the exponent does track is how far the run has progressed. The value reached at the end of each run correlates with the logarithm of its step count at $r = 0.99$, which is what would be expected if all four are approaching a common asymptote from below and have simply gone different distances towards it. On that reading the phase-field exponent of 0.259 is not a regime-specific value at all but a transient one, and the asymptote — which may well be 1/3 — lies beyond the reach of these runs.

The crossover hypothesis is therefore not supported, and the discrepancy of Section 8 returns to the explanation offered first there: that the end-member comparison pairs cooling rates with textures reported from different samples, and that paired measurements on single specimens are what would settle it.

10 Limitations

We set these out in the order in which they bear on the conclusions, and distinguish throughout between limitations that affect the numbers and limitations that would affect the argument.

The phase-field interface is artificially widened. This is stated in Section 2.5 before any phase-field result is presented, and it is absolute: the domain sizes in Figures 1 and 2 are set by the grid spacing, not predicted. Inflating κ rescales lengths and times together, so the dimensionless conclusions — the order of morphologies, the existence of arrest, the sign and approximate magnitude of the size–rate exponent — survive; the absolute scale does not, and comes instead from Section 6.

The lattice dynamics has no vacancy mechanism. Solute moves by direct exchange, which conserves composition and satisfies detailed balance but is not how atoms move in a real crystal. Indium in particular is expected to migrate only with a vacancy or a compensating counter-ion, as Section 8 discusses, and a mechanism with a different correlation factor need not give the same coarsening exponent. This is a plausible contributor to the 0.150 measured in Section 6 and is the reason that value is offered as a lower bound rather than a determination.

The simulated compositions are far above natural tenors. The lattice coarsening runs use 4 at.% total solute, some two to three orders of magnitude above the indium content of economically relevant sphalerite, because at natural dilution the solute budget is exhausted into a single domain before any coarsening can be measured — which is itself the result of Yu et al. (2026b). The exponent is a property of the transport-limited kinetics rather than of the concentration, but the domains at 4 at.% are not dilute, and interactions between them are a further reason the asymptotic law need not hold.

Monte Carlo sweeps are not physical time, and the mobility is not absolute. Section 2.6 sets out the consequence: results are relative calibrations. A ratio of domain sizes maps onto a ratio of cooling rates; neither maps onto an absolute value without a measured diffusion coefficient for Cu or In in sphalerite.

The coarsening exponent is not settled. Four determinations span a factor of three (Section 7). Section 9 suggests they may describe different regimes rather than disagreeing about one, but that suggestion is untested. Until it is, the inversion supports ranking and not calibration.

The end-member comparison is not paired. Table 4 sets deposit-class cooling rates against inclusion textures reported from different samples. Paired measurements — inclusion size distributions and independent thermal-history constraints on the same specimens — would test the power law properly and do not appear to exist. This is the single most useful observation the framework calls for.

Iron is absent, and the chemical term is assumed. Both are inherited from Yu et al. (2026a) and discussed there. Fe^{2+} is isovalent with Zn^{2+} and so does not enter the charge-compensation term, but modifies the dielectric response that fixes λ; the chemical term $J_{\alpha\beta}$ remains unconstrained by first-principles calculation, and enters γ explicitly through Eq. (8).

What none of these affects. The existence of kinetic arrest does not depend on any of the above. It follows from two facts that are not in question: that solute transport is thermally activated, so the coarsening time grows exponentially as temperature falls; and that cooling proceeds at a rate which does not. Two such functions cross, and they cross once. Every quantity that might be wrong — the diffusion prefactor, the activation energy, the interfacial energy, the domain size — enters the arrest condition of Eq. (12) inside a logarithm, and Eq. (13) shows that an order of magnitude error in any of them shifts the arrest temperature by less than a quarter.

What is uncertain is therefore the calibration, not the mechanism. The inclusions in natural sphalerite are frozen coarsening products; how precisely their size dates the freezing is what remains to be settled.

Data and code availability

Source code, numerical results and figures for this study are openly available at `https://github.com/Ruqing1963/sphalerite-kinetic-arrest` and archived at `https://doi.org/10.5281/zenodo.21995641`. Code is released under the MIT licence and data under CC BY 4.0; the manuscript remains the copyright of the authors. The repository includes the Cahn–Hilliard solver with Arrhenius mobility used in Section 5, the direct lattice coarsening of Section 6, the thermodynamic integration that measures $f(c)$ in Section 2.2, and the scripts that regenerate each figure.

Two components are released although they were not run for this work. The structure generation and least-squares harness for the first-principles determination of $J_{\alpha\beta}$ is complete and validated against synthetic energies, so it can be run by anyone with sufficient memory and the parameters dropped into Eq. (8) without repeating any simulation reported here. And the verification suite of Yu et al. (2026a) is included unchanged, since every result in this paper depends on the lattice module passing it.

The two preceding papers of the series are archived separately, with their own code: Yu et al. (2026a) at `https://doi.org/10.5281/zenodo.21880502` and Yu et al. (2026b) at `https://doi.org/10.5281/zenodo.21911898`.

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